

Supply Chain Management UI/UX Product Standard

Domain Standard #10 · Modern Supply Chain Control Tower

Design direction: Supply Chain Control Tower · Network Visibility · Predictive Planning · Exception-First Operations

Purpose: Establish a reusable UI/UX standard for demand planning, supply planning, sourcing, inventory, fulfillment, logistics, supplier collaboration, risk management and end-to-end supply chain analytics.

Modernity rule: Supply chain software should feel like a connected, intelligent control tower—not a collection of disconnected operational modules.

1. Supply Chain Management Product Definition

Supply Chain Management products coordinate demand, supply, suppliers, inventory, production and movement across a network. UX must help users understand the end-to-end flow and intervene when risks or exceptions emerge.

Core UX objective: create a shared operational picture of the supply network and guide users from signal → impact → decision → action.

Primary user groups

- **Supply planners:** demand, supply, capacity and replenishment planning.
- **Procurement teams:** supplier sourcing, purchase commitments and supplier coordination.
- **Operations teams:** inventory, production, fulfillment and exception management.
- **Logistics teams:** shipments, transportation and delivery visibility.
- **Supply chain managers:** network health, service levels, cost and risk.
- **Executives:** end-to-end performance, working capital, resilience and strategic planning.

2. Modern Supply Chain Design Language

Style type: Modern Supply Chain Control Tower / Network Intelligence Workspace

- **Visual tone:** analytical, operational, calm and decision-oriented.
- **Layout:** control-tower dashboards combined with deep operational workspaces.
- **Density:** high for planners; progressively simplified for management and executive views.
- **Hierarchy:** demand/supply balance, service impact, risk and exceptions receive priority.
- **Visualization:** network maps, timelines, forecasts and scenario views where they improve decisions.
- **Interaction:** drill-down, scenario exploration, contextual side panels, saved views and bulk exception resolution.
- **Collaboration:** preserve comments, ownership, decisions and action history around supply events.

3. Current Supply Chain UI Pattern Expectations

Pattern	Supply chain expectation	Example
Control tower	Single network view of demand, supply, inventory, logistics and exceptions.	Service risk across 4 regions.
Exception-first UX	Surface events requiring intervention before healthy operations.	Supplier delay threatens customer order.
End-to-end drill-down	Move from network KPI to the transaction causing the issue.	OTIF → region → shipment → order.
Demand/supply balance	Compare demand signal with available and expected supply.	Forecast 12,000 vs supply 9,500.
Scenario planning	Model changes without immediately altering live plans.	Supplier delay +7 days scenario.
Timeline view	Show dependencies and milestones across supply events.	PO → production → shipment → receipt.
Network visualization	Represent nodes and movement when spatial context helps.	Supplier → plant → DC → customer.

Pattern	Supply chain expectation	Example
Risk cockpit	Prioritize supply risks by impact, likelihood and urgency.	Critical supplier disruption.
Collaborative workspace	Attach owners, comments and decisions to events.	Planner assigns mitigation task.
AI assistance	Support forecasting, risk detection and recommendations with explanations.	Predicted stockout risk in 14 days.

4. Supply Chain Information Architecture

- **Control Tower:** network health, service, cost, risk and exceptions.
- **Demand Planning:** forecasts, demand signals, seasonality and scenarios.
- **Supply Planning:** supply plans, capacity, constraints and shortages.
- **Sourcing & Suppliers:** suppliers, contracts, performance and risk.
- **Procurement:** purchase commitments and inbound supply.
- **Inventory:** network stock, allocation, replenishment and aging.
- **Production:** plans, materials, capacity and production constraints where applicable.
- **Logistics:** shipments, transportation milestones and delivery exceptions.
- **Orders & Fulfillment:** demand commitments and service status.
- **Risk & Resilience:** disruptions, dependencies, scenarios and mitigation.
- **Analytics:** service, cost, working capital, forecast accuracy and network performance.

5. Supply Chain Navigation Standard

Recommended model: Control Tower → Signal → Network Context → Impact → Scenario/Decision → Action → Outcome.

- Make time horizon, region and network scope persistent.
- Provide global search across orders, SKUs, suppliers, shipments and locations.
- Preserve cross-module relationships between demand, supply, inventory and logistics.
- Use role-based workspaces for planners, procurement, logistics and management.
- Support saved views for recurring exceptions and planning cycles.
- Make ownership and next action visible for every major exception.

6. Supply Chain Design System Foundations

Foundation	Supply chain standard
Grid	12-column control-tower and planning workspace.
Spacing	4px base; 8/12/16/24/32px rhythm.
Typography	Clear hierarchy for KPIs, forecasts, quantities, dates and risk.
Numbers	Consistent units, quantities, currency, percentages and time periods.
Tables	High-density planning tables with filtering, pinned dimensions and configurable columns.

Foundation	Supply chain standard
Charts	Forecast, variance, trend and scenario visualizations with explicit definitions.
Status	Explicit semantic labels; color is supplementary.
Motion	Functional transitions for live updates, loading and scenario changes.

7. Supply Chain Control Tower

- Start with service health, supply-demand balance, inventory exposure and critical exceptions.
- Use role-aware dashboards so planners see actionable detail while executives see network-level signals.
- Show scope explicitly: region, business unit, product category and time horizon.
- Provide drill-down from KPI to affected orders, SKUs, suppliers or shipments.
- Use an exception queue with owner, impact, deadline and next action.
- Support refresh/live-state indicators when underlying data is continuously updated.

8. Demand Planning UX

- Show historical demand, forecast and actuals distinctly.
- Make forecast horizon and granularity explicit.
- Surface forecast accuracy and significant deviations.
- Support planner adjustments with clear before/after values.
- Provide scenario modeling without overwriting the baseline.
- Allow users to inspect demand drivers and supporting signals.
- Use collaborative comments and approval history for major planning changes.

9. Supply Planning UX

- Compare required supply against available and planned supply.
- Surface capacity, material and supplier constraints.
- Show shortage impact on customer service or downstream operations.
- Provide alternative supply options where available.
- Use timeline or planning-board views for dependencies.
- Make proposed changes distinguishable from committed plans.

10. Supplier & Network Risk UX

- Show supplier performance, dependency and disruption signals.
- Separate observed facts from predictions or risk scores.
- Prioritize risks by operational impact and urgency.
- Provide mitigation actions and accountable owners.
- Link risk events to affected SKUs, orders, locations or customers.
- Preserve incident and mitigation history.

11. Inventory Network UX

- Show inventory by node and network position.
- Separate available, reserved, inbound and blocked inventory.
- Highlight imbalances between locations.
- Support transfer and allocation decisions with destination/source context.
- Provide inventory aging and working-capital views.
- Connect inventory exceptions to demand and supply signals.

12. Logistics Visibility UX

- Show shipment milestones in a consistent timeline.
- Separate planned, estimated and actual timestamps.
- Surface late or at-risk shipments.
- Connect shipment exceptions to affected orders and inventory.
- Provide carrier, route and destination context where applicable.
- Use maps only when geographic movement materially improves decision-making.

13. Scenario Planning & What-If UX

Scenario planning should help users test decisions without confusing simulations with live operational data.

- Clearly label baseline versus scenario values.
- Show assumptions before showing scenario outcomes.
- Compare scenarios side-by-side.
- Highlight service, cost, inventory and capacity impacts.
- Provide save/share/approve mechanisms appropriate to the organization.
- Never silently publish a scenario into the live plan.

14. Supply Chain Analytics

- Track OTIF/service level, forecast accuracy, inventory turns, fill rate, lead time and cost.
- Use explicit metric definitions and time periods.
- Allow network → region → node → SKU/order drill-down.
- Separate actual, planned, forecast and scenario values.
- Use variance views to explain why performance changed.
- Provide export with clear scope and filters.

15. AI & Intelligent Supply Chain Standard

AI can help planners identify patterns and risks, but recommendations must remain explainable and distinguishable from committed plans.

- Use AI for demand forecasting, anomaly detection, ETA prediction, supplier-risk signals and planning recommendations.

- Show contributing signals or assumptions where feasible.
- Clearly label predicted, recommended and actual values.
- Require human review for consequential planning or supplier decisions unless explicitly authorized.
- Allow planners to override recommendations and capture rationale.
- Never silently modify committed supply plans.

16. Example — Supply Disruption Response

- **1. Detect:** control tower identifies supplier delay.
- **2. Assess:** system shows affected SKUs, orders, locations and service impact.
- **3. Explore:** planner compares alternative suppliers, transfers or inventory allocations.
- **4. Simulate:** scenario view estimates cost and service effects.
- **5. Decide:** authorized user selects mitigation.
- **6. Assign:** actions receive owners and deadlines.
- **7. Monitor:** control tower tracks recovery and outcome.

17. Example — Demand-to-Supply Planning

- Forecast demand by product and period.
- Compare demand against inventory and incoming supply.
- Identify future shortages or excess.
- Evaluate replenishment, supplier or allocation options.
- Review scenario impacts.
- Publish approved planning changes with traceability.

18. Collaboration & Decision History

- Attach comments, decisions and action owners to supply events.
- Show who changed a plan and why.
- Use mentions/tasks for cross-functional collaboration where appropriate.
- Keep decision history accessible from the affected record.
- Separate discussion from authoritative plan values.

19. Accessibility & Responsive Strategy

- Target WCAG 2.2 AA for core workflows.
- Do not communicate risk or performance using color alone.
- Provide accessible data alternatives for maps and charts.
- Support keyboard navigation and visible focus.
- **Desktop:** primary environment for planning, control towers and analytics.
- **Tablet:** exception review, approvals and operational monitoring.
- **Mobile:** alerts, exception status, approvals and action updates.
- Transform wide planning tables into prioritized views on smaller screens rather than shrinking content.

20. Supply Chain UX Quality Gates

Gate	Validation question
Network visibility	Can users understand the current end-to-end supply situation?

Gate	Validation question
Exception clarity	Can users identify what requires intervention and why?
Impact clarity	Can affected orders, SKUs, locations or customers be traced?
Planning safety	Are baseline, forecast and scenario values clearly separated?
Decision support	Does the interface provide enough context for action?
Collaboration	Are ownership, decisions and rationale visible?
Scalability	Can the product support large networks and planning datasets?
Modernity	Does the experience behave like a connected control tower rather than disconnected modules?

21. Design-Glanza Product Builder Rules

- Generate a supply-chain-specific design system before generating modules.
- Treat Demand Signal, Forecast, Supply Plan, Supplier, Purchase Order, Inventory Node, Shipment, Customer Order, Exception, Risk Event and Scenario as reusable domain objects.
- Generate realistic states such as Planned, Forecasted, Confirmed, In Transit, At Risk, Delayed, Received, Completed and Cancelled.
- Use modern patterns: control tower, exception cockpit, timeline views, network visualization, scenario comparison, saved views and intelligent recommendations.
- Keep actual, planned, forecast and scenario values visually distinct.
- Make ownership and next action first-class parts of exception design.
- Keep AI predictions and recommendations visibly separate from committed operational records.
- Audit planning, disruption, scenario, logistics and accessibility flows before final delivery.

22. Supply Chain Management Standard Summary

Style: Modern Supply Chain Control Tower / Network Intelligence Workspace

UX model: Detect → Understand → Assess Impact → Simulate → Decide → Act → Monitor

Visual priority: network state → exception → impact → scenario → decision → outcome

Modern patterns: control tower, exception-first UX, network views, scenario planning, timeline dependencies and responsible AI

Non-negotiables: end-to-end visibility, decision clarity, traceability, collaboration, planning safety and scalability.

Domain Standard #10 for the Design-Glanza domain library. Supply chain models, planning horizons, industry constraints, network structures and local operating practices should extend this baseline.